Isaac Marval Hernández

Software Development · Cybersecurity · Applied AI

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SUMMARY

Software engineer (Information Systems Engineering) with hands-on cybersecurity experience. Builds applications end to end in Python, Java and SQL, from REST APIs and databases to applied AI systems, with a background in cyber threat intelligence (OSINT, OPSEC) and systems administration. Currently building a secure, on-premise LLM system for confidential document analysis.

EDUCATION

BSc in Information Systems Engineering — Universidad de Alcalá (UAH), Madrid *(final year, thesis in progress)* 2022 – 2027
Higher Technician in Telecom & Computer Systems — Salesianos de Atocha, Madrid *(8.81/10, top of class)* 2020 – 2022

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, JavaScript, MATLAB, Bash, PowerShell, HTML, CSS
- **Software Engineering:** OOP, UML, REST APIs (FastAPI), full-stack web development, concurrent & distributed programming (Java threads, RMI), unit & black-box testing, Git, GitHub, Jira
- **Data:** PostgreSQL, relational data modeling, Power BI (data cleaning, dashboards), NumPy
- **AI & LLMs:** Large Language Models (LLM), Generative AI, Retrieval-Augmented Generation (RAG), hybrid search (BM25 + vector retrieval), embeddings, FAISS, on-premise LLM deployment, prompt engineering, machine learning (k-NN, SVM, neural networks, k-means), evolutionary algorithms, computer vision
- **AI-Assisted Development:** Claude Code (subagents, hooks, MCP, skills), GitHub Copilot, Codex
- **Security:** OSINT, cyber threat intelligence, OPSEC (sock puppets, Tails, compartmentalized identities), forensic acquisition (disk imaging, hash verification), network security fundamentals, cryptography fundamentals, Tor, Wireshark, Nmap (basic), ENS / ISO 27001
- **Systems & Networking:** Linux, Windows, Docker, virtualization (VM provisioning, resource allocation), server administration, VoIP, TCP/IP, routing & switching

WORK EXPERIENCE

Cyber Threat Intelligence Intern (OSINT) — Qintegra, Madrid Jan 2022 – Jun 2022

- OSINT monitoring of the technology sector and production of strategic intelligence reports.
- Created and maintained sock-puppet research personas under strict OPSEC: dedicated hardware, Tails (amnesic OS running from RAM), prepaid SIMs and segregated networks; trained in forensically sound data acquisition.

IT Technician (Internship) — OpenSpring S.L. / Datatronics, Madrid Jun 2022 – Sep 2022

- Deployed and administered VMs and servers per client specifications (incl. VoIP); technical support for corporate systems and networks.

TECHNICAL PROJECTS

All projects in one place: marvali.github.io

Final Thesis (TFG): Secure On-Premise LLM for Confidential Document Analysis 2026 · in progress
Self-hosted LLM with hybrid RAG (BM25 + dense retrieval) so organizations can query internal documents without data leaving their infrastructure; offline inference, document-level access control, audit logging and prompt-injection / data-leakage testing (ENS / ISO 27001).

Local RAG for F1 Regulations (GitHub) — PDF parsing, embeddings, FAISS and a local LLM in Python. Mar 2026
Recipe Social Network (GitHub) — full-stack app: FastAPI REST API, PostgreSQL, HTML/CSS/JS. 2025/26
Cookie Factory Simulation (GitHub) — concurrent Java (threads, synchronization), RMI client/server, Swing.
Multi-Object Pedestrian Tracking (GitHub) — MATLAB, YOLOv4, Kalman filter, Hungarian matching. Jan 2026
Wind Farm Layout Optimization (GitHub) — Coral Reef Optimization in Python, Numba + joblib. 2025/26
Soft-Computing Lab Series — genetic algorithms, evolution strategies, k-NN, SVMs, neural networks, k-means. Sep–Nov 2025
Post-Quantum Cryptography Demo — Kyber-512 (ML-KEM) vs. RSA-2048 benchmark in Python. Apr 2026
Tor & Dark Web OPSEC Exercise — limits of Tor anonymity, controlled security-testing proof of concept. Feb 2026
OSINT Digital Footprint Investigation — subject's routine reconstructed from public sources, ethically. 2025/26

CERTIFICATIONS & LANGUAGES

Training: Cisco NetAcad — CCNA (Switching, Routing & Wireless Essentials), CCNAv7, IT Essentials; Power BI basic course.
Languages: Spanish — native; English — C1.